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# Surgical vs transcatheter aortic valve replacement in bicuspid aortic valve stenosis: A systematic review and meta-analysis ☆,☆☆,★

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A B S T R A C T

This systematic review and meta-analysis aim to provide a comprehensive analysis of the literature directly comparing the outcomes of surgical aortic valve replacement (SAVR) and TAVR in patients with BAV stenosis. Medline, PubMed, and Scopus were systematically searched for articles published between 2000 and 2023, 1862 studies were screened, and 6 retrospective studies met the inclusion criteria. We included 6550 patients in the final analyses: 3,292 and 3,258 in the SAVR and TAVR groups, respectively. Both groups have similar rates of in-hospital mortality (odds ratio (OR) 1.11; 95% CI 0.59–2.10; $p = 0.75$) and stroke (OR 1.25; 95% CI 0.85–1.86; $p = 0.26$. Patients who underwent SAVR experienced lower rates of permanent pacemaker implantation (OR 0.54; 95% CI 0.35–0.83; $p = 0.005$) and paravalvular leak (OR 0.47; 95% CI 0.26–0.86; $p = 0.02$). On the other hand, patients who underwent TAVR displayed lower rates of acute kidney injury (OR 1.81; 95% CI 1.15–2.84; $p = 0.010$), major bleeding (OR 3.76; 95% CI 2.18–6.49; $p < 0.00001$), and pulmonary complications (OR 7.68; 95% CI 1.21–48.84; $p = 0.03$). Despite the early mortality data suggesting that TAVR may be a reasonable strategy for patients with bicuspid AS with low to intermediate surgical risk, the increased risk of PPI and PVL is concerning. A prospective, randomized, controlled trial reporting long-term outcomes with pre-defined subgroup analyses based on BAV morphology is paramount. In the interim, caution should be exercised in the widespread adoption of TAVR in lower surgical-risk patients.

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## Introduction

A bicuspid aortic valve (BAV) is the most common congenital cardiac condition, affecting 1–2% of the population [1]. Altered aortic root hemodynamics and greater mechanical stress imparted on the valve leaflets accelerate routine wear and tear, inducing earlier leaflet calcification and earlier-onset aortic stenosis (AS) [2]. Therefore, up to 50% of individuals with a BAV require aortic valve replacement (AVR) in their lifetime [1,3]. BAV stenosis is the most common cause of AS in individuals under the age of 70, and the number of elderly patients with bicuspid AS is also increasing. A BAV not only accelerates disease progression and significantly increases the risk of AS, but its unique anatomical characteristics, commonly associated with aortopathy, and additional cardiac anomalies also pose challenges to intervention [4,5].

Classically, the gold standard treatment for AS has been surgical aortic valve replacement (SAVR). However, the emergence of transcatheter technologies over the last two decades has led to debate over the optimum management strategy in patients with AS. Based on the results of early randomized controlled trials, TAVR was initially only utilized in patients of high surgical risk in whom SAVR was contraindicated [6,7]. Subsequent trials indicated non-inferiority of TAVR compared to SAVR in intermediate-risk patients [8,9]. The PARTNER III trial reported a reduced rate of a composite of death, stroke, or rehospitalization at 1 year with TAVR in low-risk patients [10]. Similarly, the Evolute low-risk trial showed that TAVR was non-inferior to SAVR in terms of the composite end point of death disabling stroke at 2 years [11]. Imperatively, patients with BAV stenosis were excluded from the landmark TAVR trials due to concerns over technical challenges with valve implantation. In addition to the frequent concomitant aortopathy, more significant leaflet and annular calcification in bicuspid AS theoretically increases the risk of annular rupture, cardiac conduction abnormalities, and paravalvular regurgitation with TAVR [4,6,12–15].

Based on the 2020 ACC/AHA guidelines for the management of patients with valvular heart disease, stage D1, D2, and D3 aortic stenosis, as well as stage C2 with LVEF < 50%, are class I indications for TAVR in patients >65 years of age [16]. Specific to BAV, the guideline lists TAVI as a potential alternative intervention op-

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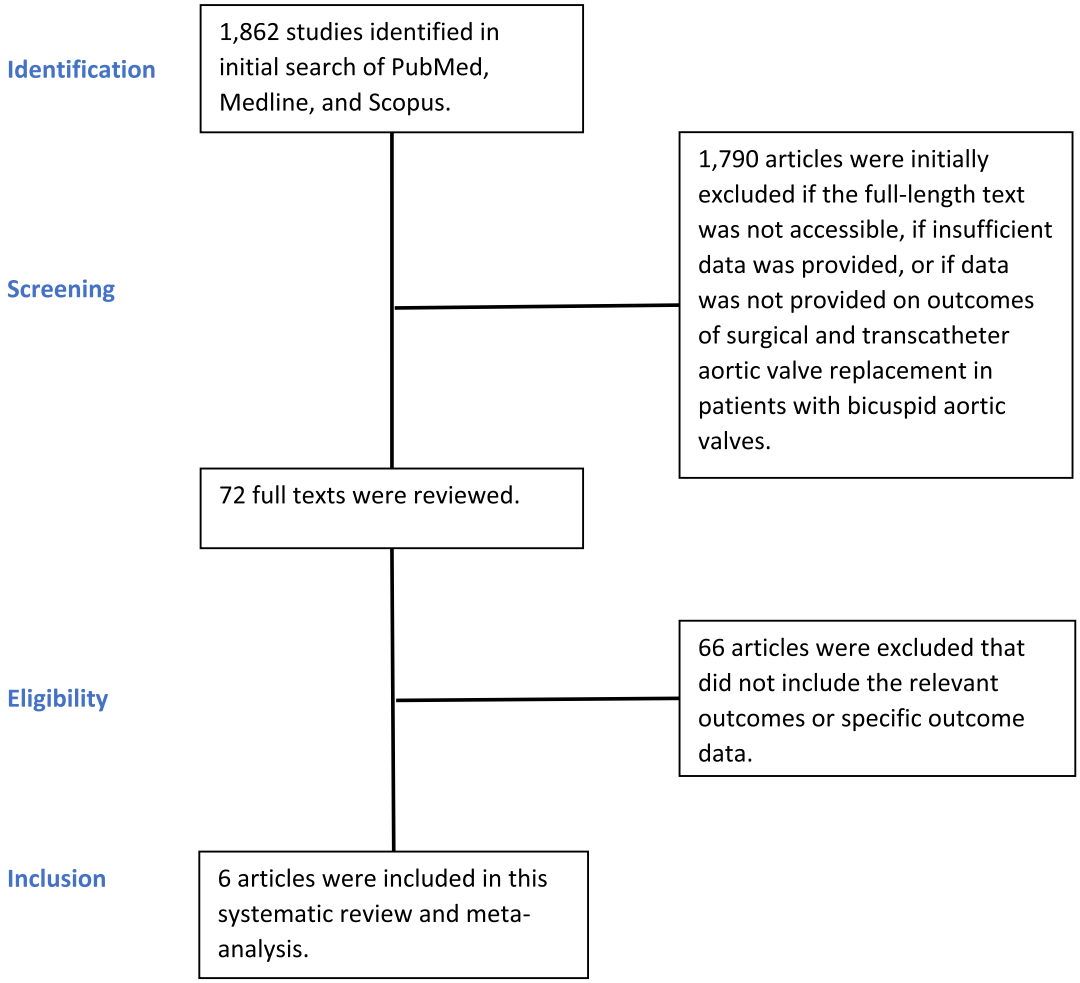


**Fig. 1.** Study flowchart based on PRISMA guidelines.

tion to SAVR as class 2b. This is after consideration of patient-specific procedural risks, values, trade-offs, and preferences, and when the surgery is performed at a comprehensive valve center [16]. The 2021 ESC/EACTS guideline lists TAVI as class 1 indication in older patients (≥75), or in those who are high risk (STS-PROM/EuroScore II>8%) or unsuitable for surgery. Specific to BAV, the guideline notes that SAVR remains more appropriate in patient with AS affecting bicuspid valve and in those with associated disease (e.g. aortic root dilatation, complex coronary disease, or severe mitral regurgitation) requiring a surgical approach) [17]. However, with significant advancements in valve and deployment technology, as well as physician expertise, TAVR in patients with BAV is gaining popularity [18]. This is occurring despite a lack of evidence beyond retrospective, observational studies with limited sample sizes [19–25].

To date, there lacks a prospective, randomized controlled trial comparing SAVR and TAVR in patients with BAV. Consequently, there is controversy as to whether SAVR or TAVR is the optimal treatment for BAV stenosis. To ensure the optimization of patient outcomes, a higher-powered analysis of the current literature is required prior to the more widespread adoption of this management strategy for BAV stenosis. The objective of this systematic review and meta-analysis is to provide a comprehensive analysis of the literature directly comparing the outcomes of SAVR and TAVR in patients with BAV stenosis.

## Methods

### *Data sources*

Medline, PubMed, and Scopus were systematically searched for articles published between January 1st, 2000, and February 10th, 2023. The reference lists from all included studies were also examined. The literature review was conducted by two authors who identified appropriate articles and extracted data using prespecified outcomes. This systematic review and meta-analysis were not registered prior to the commencement of data collection.

### *Study selection*

The updated Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were adhered to in this review [26–28]. Inclusion criteria consisted of retrospective, prospective, and randomized controlled trials reporting quantitative outcomes on SAVR and TAVR in patients with BAV. In an effort to provide a comprehensive review and analysis of the literature, a date of January 1st, 2000 was selected as the cut-off for article inclusion. Case reports, published abstracts, expert opinions, editorials, and previous reviews were excluded. Screened articles were also excluded if quantitative data on the primary or secondary outcomes were not reported, databases were overlapping, the full-

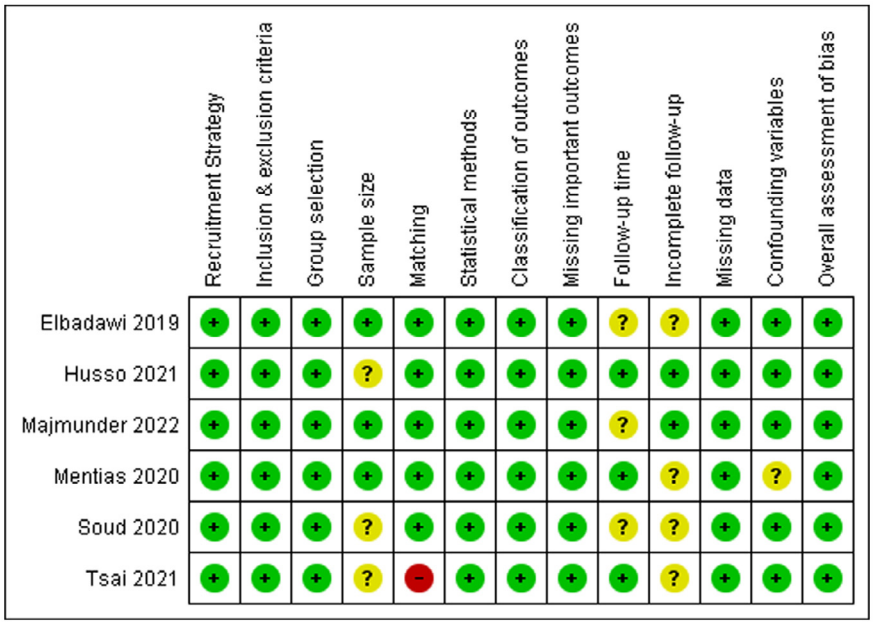


**Fig. 2.** Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

length text was not accessible, or the text was not published in English. A total of 1862 studies were screened. After extensive review, 6 studies adjudged to meet the inclusion criteria were included in the final analysis (Fig. 1). Conflict over text inclusion was resolved by the involvement of a third author and deliberation until a unanimous decision was reached.

*Data extraction*

The primary outcome of interest was early and mid-mortality, as defined by mortality rates in-hospital, as well as at 30 days, 1 year, and 2 years. Secondary outcomes included rates of postoperative morbidity; specifically myocardial infarction (MI), stroke, arrhythmias/heart block, permanent pacemaker implantation (PPI), paravalvular leak (PVL), acute kidney injury (AKI), postoperative bleeding requiring blood transfusion, respiratory complications, and length of hospital stay (LOS). Baseline preoperative characteristics, including patient age and sex, as well as cardiovascular risk factors such as hypertension (HTN), diabetes mellitus (DM), congestive heart failure (CHF), chronic pulmonary disease, and previous cardiac surgical procedures, were also extracted from the individual texts. If values/percentages were not explicitly reported for the above outcomes, they were calculated from the data provided in the studies. In studies reporting outcomes for propensity-matched patient populations, the outcomes of the matched populations were included in this review. Where available adjusted statistics were utilized.

*Critical appraisal*

A qualitative adjudication of the methodological and scientific quality of each study included in this review was conducted (Fig. 2). Previously published guidelines, with particular emphasis on the ROBINS-I tool guided the critical appraisal process [29–31]. Selection, detection, reporting, and publication biases were all taken into consideration.

*Statistical analysis*

Data were extracted from individual studies and pooled to perform a comparative analysis. Outcomes were compared among patients that had received either SAVR or TAVR. Rates of in-hospital mortality, stroke, AKI, postoperative bleeding requiring blood transfusion, PPI, PVL, and respiratory complications were compared. For outcomes with sufficient data, a subgroup analysis was performed with solely propensity-matched studies. A random effect model was utilized when comparisons were found to have considerable heterogeneity between studies. A fixed-effect model was used when heterogeneity was not found to be significant between studies. Odds ratio (OR) and 95% confidence intervals (CI) were reported using the Mantel-Haenszel method for comparisons without significant heterogeneity and the Inverse Variance method for comparisons with significant heterogeneity. Pooled data for comparisons were represented in forest plots. The $Chi^2$ statistic was used to identify heterogeneity between studies and the $I^2$ statistic identified degrees of difference thought to be due to effect size. Statistical significance was set at a $p < 0.05$ for all comparisons. Statistical analysis was performed, and forest plots were created using the RevMan 5.4 software (Cochrane Training, London, UK).

## Results

*Overview of included studies*

6 studies were included in this review [19–23,25]. The study population consisted of 6550 patients who underwent aortic valve replacement for bicuspid aortic stenosis; 3292 of which underwent SAVR and 3258 of which underwent TAVR. Data collection dates ranged from 2008 to 2018. Follow-up ranged from in-hospital to 6 years. Solely retrospective studies were identified in the literature and comprised the study population. The characteristics of the included studies are reported in Table 1, baseline preoperative characteristics in Table 2, mortality rates in Table 3, and rates of postoperative morbidity in Table 4. Forest plots of the meta-analysis are included in Figs. 3 and 4 and Supplementary Fig. 1.

*Quality of evidence*

All 6 studies were assessed to be at low risk of bias. The risk of bias summary was created with RevMan 5.4 software (Cochrane

**Table 1**
Characteristics of each included study.

| Study name | Study Type | Number of Patients Analyzed | Years Data Collected | Follow-up Period | Study Population | Propensity-Matching |
|---|---|---|---|---|---|---|
| Elbadawi 2019 | Retrospective | Matched SAVR: 975<br>Matched TAVR: 975<br>Matched Total: 1950 | 2012–2016 | In-hospital | Patients with bicuspid AS undergoing SAVR or TAVR | Yes |
| Husso 2021 | Retrospective | Matched SAVR: 75<br>Matched TAVR: 75<br>Matched Total: 150 | 2008–2017 | Mean = 4.6 ± 2.7 years<br>Median = 3.0 years | Patients with bicuspid AS undergoing SAVR or TAVR | Yes |
| Majmunder 2022 | Retrospective | Matched SAVR: 1393<br>Matched TAVR: 1393<br>Matched Total: 3761 | 2016–2018 | 6 months | Patients with bicuspid AS undergoing SAVR or TAVR | Yes |
| Mentias 2020 | Retrospective | Matched SAVR: 699<br>Matched TAVR: 699<br>Matched Total: 1398 | 2015–2017 | Median = 631 days | Patients with bicuspid AS undergoing SAVR or TAVR | Yes |
| Soud 2020 | Retrospective | Matched SAVR: 68<br>Matched TAVR: 68<br>Matched Total: 136 | 2011–2014 | In-hospital | Patients with bicuspid AS undergoing SAVR or TAVR | Yes |
| Tsai 2021 | Retrospective | SAVR: 82<br>TAVR: 48<br>Total: 130 | 2013–2018 | Mean = 2.86 ± 1.47 years | Patients with bicuspid AS undergoing SAVR or TAVR | No |

SAVR, surgical aortic valve replacement; TAVR, transcatheter aortic valve replacement; AS, aortic stenosis.

**Table 2**
Summary of preoperative characteristics of patients with bicuspid aortic valve stenosis undergoing SAVR or TAVR.

| Study name | Group | Age | Female | Hypertension | Diabetes | CHF (NYHA III/IV) | Chronic Pulmonary Disease | Previous Cardiac Surgery |
|---|---|---|---|---|---|---|---|---|
| Elbadawi 2019 | Matched SAVR (%) | 65.20 ± 11.52 | 355 (36.4) | 630 (64.6) | 300 (30.8) | 60 (6.2) | 250 (25.6) | 85 (8.7) |
| | Matched TAVR (%) p | 65.70 ± 16.53 | 390 (40.0) | 630 (64.6) | 290 (29.7) | 45 (4.6) | 315 (32.3) | 100 (10.3) |
| Husso 2021 | Matched SAVR (%) | 75.7 ± 6.3 | 34 (45.3) | | 11 (14.7) | 45 (60.0) | 19 (25.3) | 6 (8.0) |
| | Matched TAVR (%) p | 75.8 ± 8.4 | 33 (44.0) | | 16 (21.3) | 56 (74.7) | 16 (21.3) | 4 (5.3) |
| Majmunder 2022 | Matched SAVR (%) | 68.1 ± 8.6 | 535 (38.4) | 1053 (75.6) | 410 (29.4) | 920 (66.0) | 400 (28.7) | |
| | Matched TAVR (%) p | 68.3 ± 10.1 | 526 (37.8) | 1098 (78.8) | 391 (28.0) | 935 (67.1) | 436 (31.3) | |
| Soud 2020 | Matched SAVR (%) | 64.6 ± 12.4 | 19 (27.9) | 36.8 | 23.5 | (4.4) | 23.5 | |
| | Matched TAVR (%) p | 65.0 ± 14.8 | 22 (32.4) | 72.1 | 23.5 | (0.0) | 25.0 | |
| Tsai 2021 | SAVR (%) | 51.5 ± 32.3 | 25 (30.5) | 40 (48.8) | 10 (12.2) | 18 (22.0) | | 0 (0.0) |
| | TAVR (%) p | 65.7 ± 7.8 | 20 (41.7) | 29 (60.4) | 16 (33.3)<br><0.05 | 15 (31.3) | | 1 (2.1) |

SAVR, surgical aortic valve replacement; TAVR, transcatheter aortic valve replacement; CHF, congestive heart failure; NYHA, New York Heart Association.

**Table 3**
Summary of mortality rates for patients with bicuspid aortic valve stenosis undergoing SAVR or TAVR.

| Study name | Group | In-hospital | 30-day | 1-year | 2-year | Other Follow-up |
|---|---|---|---|---|---|---|
| Elbadawi 2019 | Matched SAVR (%) | 30 (3.1) | | | | |
| | Matched TAVR (%) p | 30 (3.1)<br>>0.999 | | | | |
| Husso 2021 | SAVR (%) | | 22 (2.4) | | 101 (11.0) | |
| | TAVR (%) p | 4 (5.3) | 1 (0.9) 0.721 | | 7 (6.5) <0.0001 | |
| | Matched SAVR (%) | 1 (1.3) | 4 (5.3) | | (18.7) | |
| | Matched TAVR (%) p | | 1 (1.3) 0.375 | | (9.7) 0.268 | |
| Majmunder 2022 | Matched SAVR (%) | 26 (1.8) | | | | |
| | Matched TAVR (%) p | 10 (0.7) | | | | |
| Mentias 2020 | Matched SAVR (%) | 16 (2.3) | 19 (2.7) | 52 (7.4) | | |
| | Matched TAVR (%) p | 15 (2.2) 0.90 | 20 (2.9) 0.90 | 63 (9.0) 0.34 | | |
| Soud 2020 | Matched SAVR (%) | 0 (0.0) | | | | |
| | Matched TAVR (%) p | 4 (5.9) 0.11 | | | | |
| Tsai 2021 | | | | | | 6-year freedom from death |
| | SAVR (%) | 4 (4.9) | | | | (56.9) |
| | TAVR (%) p | 6 (12.5) | | | | (84.2) NS |

SAVR, surgical aortic valve replacement; TAVR, transcatheter aortic valve replacement; NS, not significant.

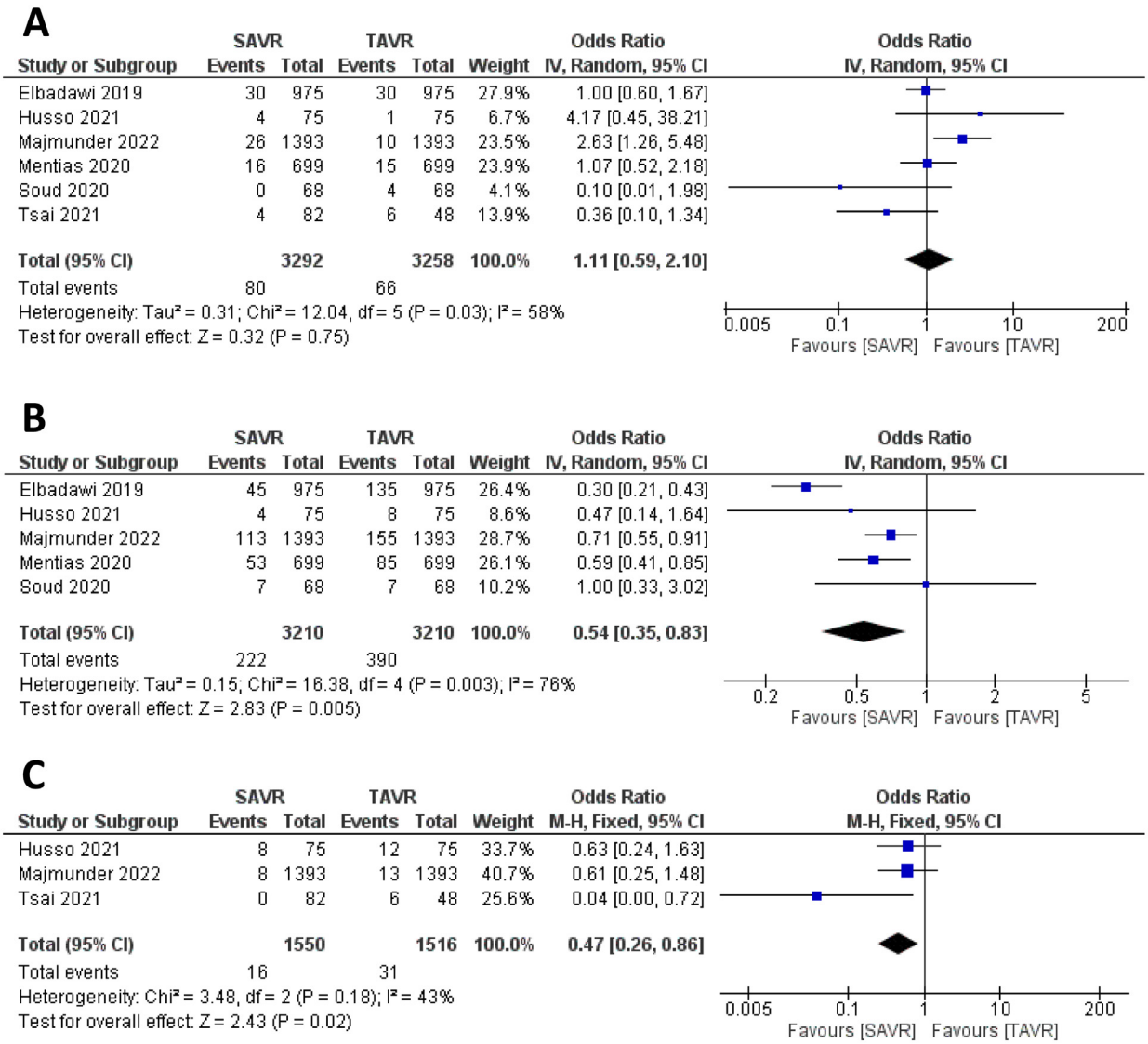


**Fig. 3.** Forest plot demonstrating in-hospital mortality (A), rates of postoperative PPI (B), and rates of postoperative PVL (C) in patients with bicuspid AS undergoing SAVR or TAVR. SAVR, surgical aortic valve replacement; TAVR, transcatheter aortic valve replacement; PPI, permanent pacemaker insertion; PVL, paravalvular leak; CI, confidence interval; df, degrees of freedom.

Training, London, UK) and using the ROBINS-I tool in combination with additional sources (Fig. 2) [29–31].

*Preoperative characteristics*

There were no differences in age, sex, or the presence of HTN, DM, NYHA III-IV CHF, chronic pulmonary disease, or previous cardiac surgical procedures between patients undergoing SAVR and TAVR in the studies with propensity-matched patient populations [19–21,25]. The sole study that did not employ propensity-matching indicated an increased prevalence of DM and a higher Society of Thoracic Surgery (STS) predicted risk of mortality in the TAVR population [22].

*Mortality*

In-hospital mortality was reported in all studies, ranging from 0.0 to 5.3% and 0.7–12.5% in the SAVR and TAVR groups, respectively [20,21,23,25]. Overall, pooled estimates indicated no differences between the groups (OR 1.11; 95% CI 0.59–2.10; $p = 0.75$). There was considerable heterogeneity between the studies in the analysis (z score = 0.32; $Chi^2$=12.04; $I^2$=58%; $p = 0.03$; Fig. 3A). A subgroup analysis of propensity-matched studies yielded similar results (OR 1.33; 95% CI 0.71–2.50; $p = 0.37$; Supplementary Fig. 1A).

2 studies reported 30-day mortality, ranging from 2.7 to 5.3% and 1.3–2.9% in the SAVR and TAVR groups, respectively [19,20]. 1-year mortality was reported in 1 study. Mortality was 7.4% in the SAVR group and 9.0% in the TAVR group [20]. 2-year mortality was also reported in 1 sole study. Mortality was 18.7% and 9.7% in the SAVR and TAVR groups, respectively, with no statistical difference between the groups [19].

*MI*

MI was noted in 2 studies, ranging from 3.1 to 4.9% and 0.0% of patients undergoing SAVR and TAVR, respectively, with no significant difference noted in individual studies [22,25].

**Table 4**
Summary of postoperative morbidity for patients with bicuspid aortic valve stenosis undergoing SAVR or TAVR.

| Study name | Group | MI | Stroke | AKI | Postoperative Bleeding | Blood Transfusions | Permanent Pacemaker Insertion (PPI) | Paravalvular leak (PVL) | Respiratory Complications | LOS (days) |
|---|---|---|---|---|---|---|---|---|---|---|
| Elbadawi 2019 | Matched SAVR (%) | 30 (3.1) | 25 (2.6) | 145 (14.9) | 435 (44.6) | 265 (27.2) | 45 (4.6) | | 35 (3.6) | 7 (5–9) |
| | Matched TAVR (%) p | NR 0.002 | 20 (2.1) 0.547 | 141 (14.4) 0.789 | 230 (23.6) <0.001 | 70 (7.2) <0.001 | 135 (13.8) <0.001 | | 20 (2.1) 0.055 | 4 (2–7) 0.001 |
| Husso 2021 | Matched SAVR (%) | | 6 (8.0) | 12 (16.2) | 16 (21.3) | 52 (69.3) | 4 (5.5) | 8 (10.7) | | 8.1 ± 4.4 |
| | Matched TAVR (%) p | | 3 (4.0) 0.508 | 2 (2.7) 0.065 | 2 (2.7) 0.001 | 9 (12.3) <0.0001 | 8 (11.4) 0.388 | 12 (16.0) 0.454 | | 4.4 ± 3.0 <0.0001 |
| Majmunder 2022 | Matched SAVR (%) | | 6 (0.7) | 282 (20.2) | 145 (10.5) | | In-hospital 113 (8.6) 155 (11.8) 0.033 | 8 (0.6) | | 6 (5–8) |
| | Matched TAVR (%) p | | 1 (0.1) 0.206 | 154 (11.1) <0.001 | 35 (2.5) <0.001 | | | 13 (0.9) 0.582 | | 2 (1–4) <0.001 |
| Mentias 2020 | Matched SAVR (%) | | 26 (3.7) | 153 (21.9) | | 150 (21.5) | 53 (7.6) | | 36 (5.2) | |
| | Matched TAVR (%) p | | 28 (4.0) 0.34 | 72 (10.3) <0.001 | | 37 (5.3) <0.001 | 85 (12.2) 0.009 | | 13 (1.9) <0.001 | |
| Soud 2020 | Matched SAVR (%) | | 0 (0.0) | 12 (17.6) | | 22 (13.2) | 7 (10.3) | | 64 (29.4) | |
| | Matched TAVR (%) p | | 1 (1.5) 1.00 | 10 (14.7) 0.81 | | 5 (7.4) 0.39 | 7 (10.3) 1.00 | | 8 (11.8) 0.01 | |
| Tsai 2021 | SAVR (%) | 4 (4.9) | 0 (0.0) | | 0 (0.0) | | | 0 (0.0) | | 17.3 ± 8.2 |
| | TAVR (%) p | 0 (0.0) NS | 1 (2.0) NS | | 0 (0.0) NS | | | 6 (12.5) NS | | 17.3 ± 14.9 NS |

SAVR, surgical aortic valve replacement; TAVR, transcatheter aortic valve replacement; MI, myocardial infarction; AKI, acute kidney injury; LOS, length of hospital stay; PPI, permanent pacemaker insertion; NR, not reportable; NS, not significant.

*PPI*

PPI was required in 4.6–10.3% of patients undergoing SAVR, as compared to 10.3–13.8% of patients undergoing TAVR [19–21,23,25]. PPI was only reported in propensity-matched studies and pooled estimates significantly favored SAVR overall (OR 0.54; 95% CI 0.35–0.83; $p = 0.005$; Fig. 3B). There was considerable heterogeneity between the included studies ($Chi^2$=16.38; $I^2$=76%; $p = 0.003$).

*PVL*

PVL rates ranged from 0.0 to 10.7% and 12.5–16.0% following SAVR and TAVR, respectively [19,22,23]. Composite analysis significantly favoured SAVR (OR 0.47; 95% CI 0.26–0.86; $p = 0.02$; Fig. 3C) with moderate heterogeneity between studies ($Chi^2$=3.48; $I^2$=43%; $p = 0.18$).

*Stroke*

Rates of stroke were 0.0–8.0% in patients undergoing SAVR and 0.1–4.0% in patients undergoing TAVR [19–23,25]. Aggregate analysis indicated no difference between the two groups (OR 1.25; 95% CI 0.85–1.86; $p = 0.26$; Fig. 4A). Heterogeneity between the studies included in the analysis was insignificant ($Chi^2$=4.86; $I^2$=0%; $p = 0.43$). Subgroup analysis of propensity-matched studies similarly indicated no difference between the interventions (OR 1.30; 95% CI 0.88–1.93; $p = 0.19$; Supplementary Fig. 1B).

*AKI*

Rates of AKI were only reported in propensity-matched studies and ranged from 14.9 to 21.9% and 2.7–14.7% in the SAVR and TAVR groups, respectively [19–21,23,25]. Pooled estimates favored TAVR (OR 1.81; 95% CI 1.15–2.84; $p = 0.010$). There was considerable heterogeneity between the studies included in the analysis ($Chi^2$=27.14; $I^2$=85%; $p < 0.0001$; Fig. 4B).

*Bleeding requiring blood transfusion*

Rates of major postoperative bleeding ranged from 0.0 to 44.6% and 0.0–23.6% following SAVR and TAVR, respectively [19,22,23,25]. Aggregate analysis favored TAVR both overall (OR 3.76; 95% CI 2.18–6.49; $p < 0.00001$; Fig. 4C) and in a subgroup analysis of propensity-matched studies (OR 3.76; 95% CI 2.18–6.49; $p < 0.00001$; Supplementary Fig. 1C). There was considerable heterogeneity between the studies included in the overall analysis ($Chi^2$=8.85; $I^2$=77%; $p = 0.01$).

*Respiratory complications*

Postoperative respiratory complications occurred in 3.6–29.4% and 1.9–11.8% of patients following SAVR and TAVR, respectively [20,21,25]. Pooled estimates favored TAVR (OR 7.68; 95% CI 1.21–48.84; $p = 0.03$; Fig. 4D). The heterogeneity between the studies in the analysis was considerable ($Chi^2$=36.63; $I^2$=95%; $p < 0.00001$).

*LOS*

3 individual studies noted significantly increased hospital LOS following SAVR as compared to TAVR [19,23,25].

## Discussion

The outcomes of this systematic review and meta-analysis have indicated several notable points. First and foremost, aggregate

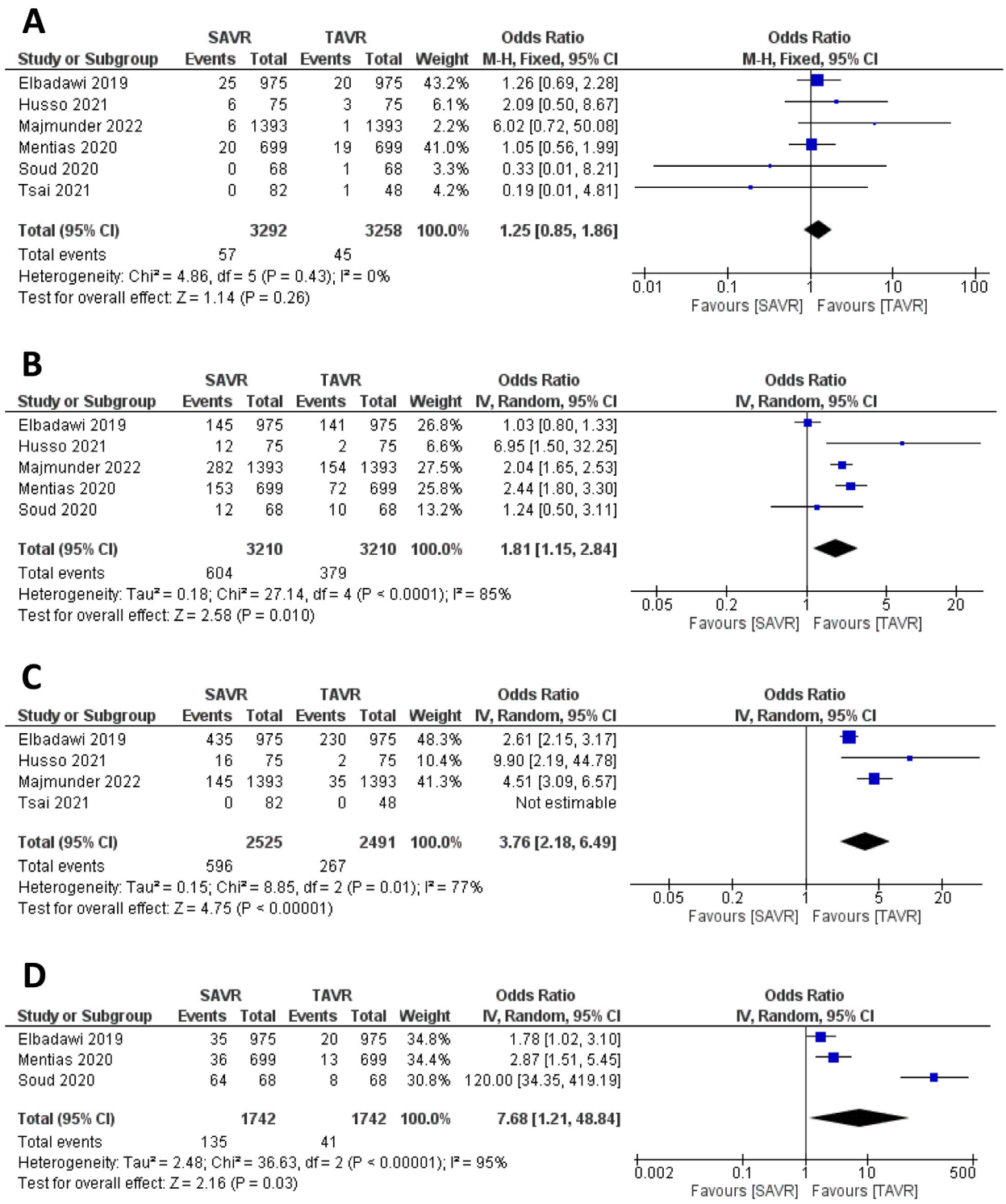


**Fig. 4.** Forest plot demonstrating rates of stroke (A), AKI (B), postoperative bleeding requiring blood transfusions (C), and respiratory complications (D) in patients with bicuspid AS undergoing SAVR or TAVR. SAVR, surgical aortic valve replacement; TAVR, transcatheter aortic valve replacement; AKI, acute kidney injury; CI, confidence interval; df, degrees of freedom.

analysis of the current literature revealed no significant difference in early mortality in patients with BAV stenosis undergoing SAVR compared to TAVR (Fig. 3A). Based on the results of individual retrospective studies, there also appeared to be no difference in mid-term mortality following SAVR and TAVR; however, sample size prohibited a pooled analysis. Regarding postoperative morbidity, composite analysis favored SAVR for rates of PPI (Fig. 3B) and PVL (Fig. 3C); and conversely favored TAVR in terms of AKI, postoperative bleeding requiring blood transfusion, and respiratory complications (Fig. 4B-D). Trends also indicated reduced hospital LOS following TAVR. There was no difference in rates of stroke between SAVR and TAVR (Fig. 4A), and sample size barred comparative analysis of postoperative MI rates. Subgroup analysis of propensity-matched studies for in-hospital mortality, as well as rates of stroke

and major bleeding yielded equivalent outcomes (Supplementary Fig. 1).

The discrepancies in postoperative morbidity reported in this meta-analysis are consistent with previous literature and are reflective of the nature of the interventions being compared. Most of the secondary outcomes in this analysis are measures of surgical morbidity and consequently would be expected to be greater following SAVR as compared to TAVR. For instance, decreased duration of intubation with SAVR compounded by less postoperative pain and analgesic use are likely to underlie the significantly increased rate of postoperative respiratory complications noted in Table 4 and Fig. 4D [21,32]. Similarly, the increased rates of blood transfusions, AKI, and postoperative LOS are consistent with the invasiveness of a surgical as opposed to a percutaneous approach. Conversely, PPI and PVL are more indicative of differences in intervention type as opposed to surgical morbidity. Reduced rates of PPI and PVL following SAVR are in keeping with anatomical concerns of TAVR in BAV stenosis [4,6,12–15]. Both outcomes have been previously noted as a potential source of apprehension in the application of this technology to BAV stenosis.

Patients with BAV stenosis were excluded from the early TAVR trials for anatomical, technical, and pathophysiological concerns. Considerably increased and asymmetric leaflet and annulus calcification theoretically pose a greater risk of improper valve seating, annular rupture, and paravalvular regurgitation. Concomitant aortopathy potentially increases the risk of aneurysmal dilation and aortic dissection [4,13,14,21]. This is all compounded by limited long-term data in a patient population undergoing intervention at a comparatively earlier stage of life. To elucidate whether some of these concerns, particularly device success, and postoperative complications, are clinically relevant, several studies have compared the outcomes of TAVR in BAV vs tricuspid aortic valve (TAV) stenosis. A recent meta-analysis by Makkar et al. reviewed the literature for the same. Although there were no significant differences in early mortality, as well as stroke or MI at the longest follow-up (1 year), moderate or severe paravalvular leak did trend towards being significantly greater in BAV stenosis (1.8% vs 1.1%, $p = 0.07$) and device success trended towards inferiority with BAV as compared to TAV stenosis (96.5% vs 97.3%, $p = 0.06$) [33].

Trends of increased paravalvular leak in the meta-analysis by Makar et al. and the present study are particularly intriguing considering recent literature analyzing outcomes of TAVR in BAV stenosis and sub-dividing patients based on BAV morphology. Husso et al. have demonstrated that specific outcomes following TAVR in bicuspid AS may differ based on BAV morphology as defined by the Sievers classification system [19,34]. The incidence of paravalvular regurgitation was significantly greater in type 1N-L and type 2L-R/R-N BAV morphologies. Consequently, the question of whether SAVR or TAVR is superior in BAV stenosis appears to be more nuanced. Specific BAV anatomy may be more amenable to successful percutaneous intervention and satisfactory long-term outcomes; however, further research with sub-analyses of outcomes based on BAV morphology is ultimately necessary.

Finally, TAVR has unquestionably advanced significantly over the last two decades and newer generation devices have shown promise in potentially mitigating select technical concerns with implantation [12,35,36]. Concurrently, SAVR continues to evolve, with minimally invasive approaches and ever-advancing valve technology reducing operative, cardiopulmonary bypass, and aortic cross-clamp times, as well as reducing postoperative pain, hastening recovery, improving cosmesis, and ultimately patient satisfaction. SAVR continues to be the standard of care when compared to minimally invasive AVR (MIAVR), whether via mini-sternotomy or right anterior thoracotomy [37–39]. However, with contemporary literature reporting favorable outcomes following MIAVR and current practice favoring minimally invasive approaches, the question of optimal treatment for BAV stenosis may soon become MIAVR vs TAVR.

Overall, although the early mortality data do tentatively suggest that TAVR is a reasonable management strategy for patients with bicuspid AS with low to intermediate surgical risk, the increased risk of PPI and PVL is concerning. A prospective, randomized controlled trial reporting long-term outcomes with subgroup analyses based on BAV morphology is ultimately required prior to more widespread adoption of this intervention.

### *Limitations*

The present systematic review and meta-analysis has certain limitations. First and foremost, the aggregate analysis is hindered by small sample size. Meta-analysis of PVL and respiratory complications was based on compiled outcomes from 3 studies. This is a consequence of the paucity of literature on the topic. While there are several observational studies analyzing outcomes of TAVR in BAV stenosis, as well as comparing the efficacy of TAVR for BAV and TAV stenosis, there is a scarcity of studies directly that directly compare SAVR and TAVR for BAV stenosis. Furthermore, given the limited guideline descriptions recommending TAVR preferentially in patients with BAV, limited data in this patient population is to be expected. In addition, the small sample size of the meta-analysis is compounded by considerable heterogeneity between individual studies and outcomes. However, we have attempted to mitigate the impact of the same in our statistical methods. Additionally, the results of this meta-analysis are limited to early and mid-term outcomes. This is reflective of the relatively recent initiation of TAVR in bicuspid AS and the consequently limited duration of follow-up.

## Conclusions

This review found that low- and intermediate-risk patients with severe stenosis of a bicuspid aortic valve had similar early mortality rates with both SAVR and TAVR. However, there were differences in the postoperative mobility between the two groups. TAVR had better outcomes in terms of renal, hemostatic, and respiratory complications, while SAVR had lower rates of PPI and PVL. Despite TAVR showing promise as a reasonable strategy in this patient group based on short-term mortality data, the increased risk of PPI and PVL is a cause for concern, especially in younger patients. A prospective, randomized controlled trial is necessary to definitively establish which intervention is superior. In the interim, caution should be exercised in the application of this intervention in this patient population.

## Author contributions

Jimmy JH Kang: Study design, data analysis, data interpretation, writing of the manuscript, proof reading and revisions, approval of final manuscript.

Nicholas M Fialka: Study design, literature search, data collection, data analysis, data interpretation, generation of figures, writing of the manuscript, approval of final manuscript.

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## Ethical statement

We the undersigned declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We confirm that all authors are responsible for the content and have read and approved the manuscript; and that the manuscript conforms to the Uniform Requirements for Manuscripts Submitted to Biomedical Journals published in *Annals in Internal Medicine*

We understand that the Corresponding Author is the sole contact for the Editorial process. He/she is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs.

Supplementary Fig. 1. Sub-group analysis of propensity-matched studies and its forest plot demonstrating in-hospital mortality (Supplementary Fig. 1A), rates of postoperative stroke (Supplementary Fig. 1B), and rates of postoperative bleeding requiring blood transfusion (Supplementary Fig. 1C).

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.tcm.2023.04.004.

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